**B. K. Birla College of Arts, Science and Commerce, Kalyan (W)**

TYBSc Physics, Semester VI  Special Theory of Relativity

# Assignment 4

Unit 2: Moving Charges, Covariance of Maxwell's Equations and Applications

Name: _____  Roll No.: _____  Date: _____

**Total marks: 60** **Each question carries 5 marks**

*Instructions:* Attempt all questions. Section A: theory (write to the point, with neat diagrams where useful). Section B: numerical problems and derivations (state formulae used, show all steps and give units). Sign convention for the interval: $s^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 - c^2\Delta t^2$; four-vectors $a^\mu = (a^0, \mathbf{a})$ with $a \cdot b = -a^0 b^0 + \mathbf{a} \cdot \mathbf{b}$.

## Section A: Theory  (6 × 5 = 30 marks)

**Q1.** Describe the electric and magnetic fields of a point charge moving with constant velocity. Explain the "pancake" shape of the field lines at high speed, and the fact that $\mathbf{E}$ points away from the *present* position of the charge. What does $\mathbf{B}$ reduce to at low speeds? **[5]**

**Q2.** Distinguish between *invariance* and *covariance*, giving examples of each. Why are Maxwell's equations covariant under the Lorentz transformation but not under the Galilean transformation? **[5]**

**Q3.** Explain why $\mathbf{E}$ and $\mathbf{B}$ cannot be the spatial parts of four-vectors, and why together they form an antisymmetric second-rank tensor. Write the field tensor $F^{\mu\nu}$ in matrix form and state how it transforms. **[5]**

**Q4.** Explain the current-density four-vector $J^\mu$. Write the continuity equation in covariant form and explain why conservation of charge holds in every inertial frame. **[5]**

**Q5.** Explain the four-potential $A^\mu$ and the Lorenz gauge condition. Write Maxwell's equations in terms of the potentials in covariant form. **[5]**

**Q6.** Discuss applications of relativistic electrodynamics: cyclotrons and synchrotrons, synchrotron radiation and relativistic beaming, and the appearance of an electromagnetic wave to a moving observer. **[5]**

## Section B: Numericals and Derivations  (6 × 5 = 30 marks)

**Q7.** An electron moves with speed $0.95c$. Find the magnitude of its electric field at a distance of $1.0\,\mu$m (a) perpendicular to its line of motion and (b) along its line of motion, and compare both with the field of an electron at rest. (c) Find the magnetic field at the point in (a). **[5]**

**Q8.** Two charges $q$ move side by side with the same velocity $v$, separated by a distance $d$ perpendicular to the motion. Using the fields of a moving charge, show that the force between them in the laboratory is $F_0/\gamma$, where $F_0$ is the Coulomb force in their rest frame. Evaluate it for two protons $1.0 \times 10^{-10}\,$m apart moving at $0.6c$. **[5]**

**Q9.** Starting from $\partial F^{\mu\nu}/\partial x^\nu = \mu_0 J^\mu$, show that the $\mu = 0$ component gives Gauss's law and the $\mu = 1$ component gives the $x$-component of the Ampère–Maxwell law. **[5]**

**Q10.** (a) Show that $\partial J^\mu/\partial x^\mu = 0$ is the equation of continuity. (b) With $J^\mu = \rho_0 \eta^\mu = \rho_0 \gamma(c, \mathbf{u})$, show that $J_\mu J^\mu = -\rho_0^2 c^2$, a Lorentz invariant. **[5]**

**Q11.** Prove that the d'Alembertian operator $\Box^2 = \nabla^2 - \dfrac{1}{c^2}\dfrac{\partial^2}{\partial t^2}$ is invariant under the Lorentz transformation. **[5]**

**Q12.** A proton of kinetic energy 500 MeV moves in a circle in a uniform magnetic field of 1.5 T. Find (a) $\gamma$ and its momentum in MeV/$c$, (b) the radius of the orbit, (c) its angular frequency of revolution. (d) Compare (b) and (c) with the non-relativistic values. **[5]**

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**Useful constants:** $c = 3.0 \times 10^8$ m/s; $e = 1.6 \times 10^{-19}$ C; $m_e c^2 = 0.511$ MeV; $m_p c^2 = 938.3$ MeV; $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m; $\mu_0 = 4\pi \times 10^{-7}$ H/m; $1/(4\pi\varepsilon_0) = 9.0 \times 10^9$ N m$^2$/C$^2$; $G = 6.67 \times 10^{-11}$ N m$^2$/kg$^2$; $M_\odot = 1.99 \times 10^{30}$ kg; 1 ly $= 9.46 \times 10^{15}$ m; 1 pc $= 3.086 \times 10^{16}$ m $= 3.26$ ly; 1 yr $= 3.156 \times 10^7$ s; $b_{\text{Wien}} = 2.898 \times 10^{-3}$ m K; $k_B = 8.62 \times 10^{-5}$ eV/K; $a = 7.566 \times 10^{-16}$ J m$^{-3}$ K$^{-4}$.